Card Balance Voice Bot

Use case: self-service credit card balance and due date by voice, with Voice ID

Domain	Retail banking, credit cards (UAE, AED)
Channel	Voice (phone call, simulated in a web page)
Language	English; Arabic planned (language is a setting)
Status	Working local demo with synthetic data only
Author / date	Sandeep Pandita, 3 November 2026
Code	github.com/sppandita85/voice-speaker-verification-banking

1. Summary

Card holders often call only to ask how much they owe and when it is due. These calls are short and routine, but each one still needs an agent to identify the caller and read figures from the card system. This use case lets a voice bot handle the whole journey: it verifies the caller by their voice, reads the outstanding balance, minimum payment and due date from the card system, and speaks the answer. No human agent is needed. If the bot cannot verify the caller, or the request is out of scope, it hands the call to an agent with a reason code.

> **Customer asks:** "How much do I owe on my credit card and when is it due?"
> **Bot answers (after Voice ID):** "The outstanding balance on your card ending 5506 is AED 4,250.75. The minimum payment of AED 212.54 is due on 18 November. Is there anything else I can help with?"

2. Goals

- Serve routine balance and due-date questions end to end, without an agent.
- Verify callers quickly and without passwords, using Voice ID, with a safe fallback.
- Never speak a wrong amount or date, and never reveal account data to an unverified caller.
- Keep a complete, privacy-safe record of every turn for audit and tuning.

3. Actors and scope

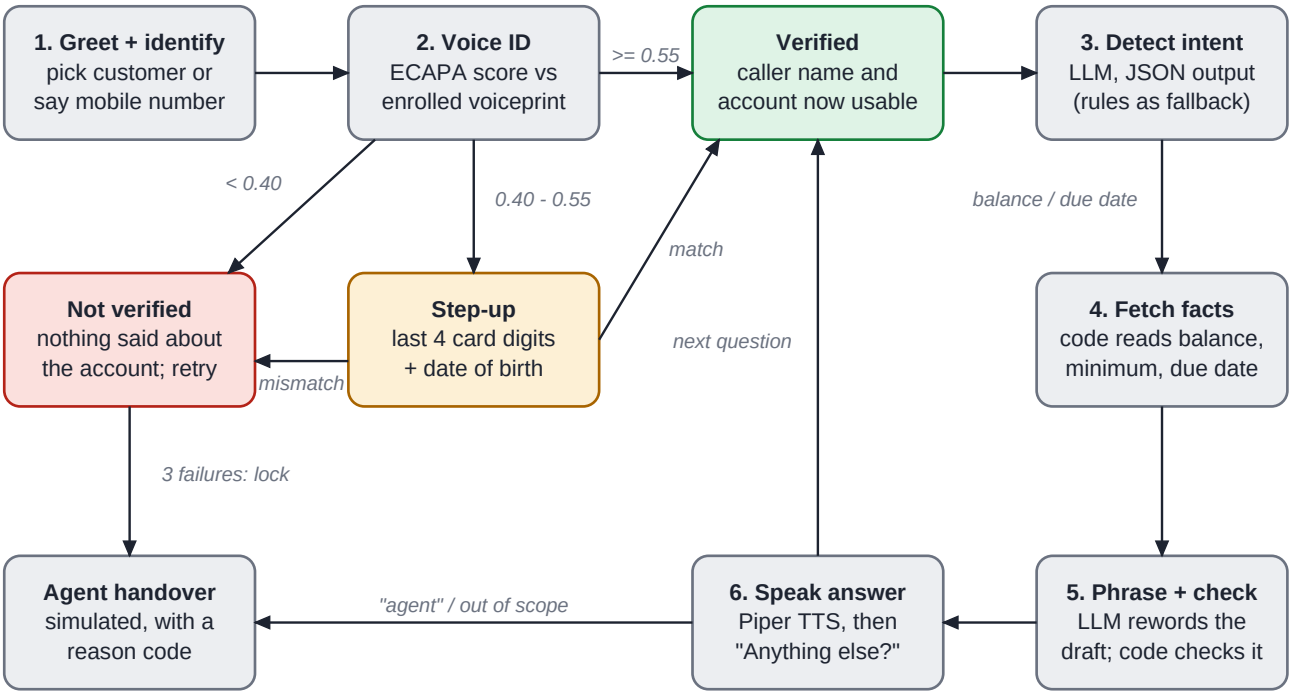
Actor	Role
Customer (caller)	Card holder who wants their balance, minimum payment or due date.
Voice bot	Identifies and verifies the caller, answers the question, offers a handover.
Contact-centre agent	Takes over when verification fails or the request is out of scope (simulated).
Card system	Source of truth for cards and statements (synthetic PostgreSQL database).

In scope	Out of scope (handed over)
Outstanding balance; minimum payment and due date; both together; choosing between two cards; overdue, blocked and zero-balance cards; repeat; goodbye; asking for an agent.	Payments, credit limit changes, lost or stolen cards, disputes, loans, rewards, and any other request.

Preconditions

- The customer is registered with a mobile number and has at least one credit card.
- For Voice ID, the customer has enrolled a voiceprint (3 short recorded sentences). Customers without one go straight to the step-up check.

4. Call flow and main success scenario



A question asked before verification is remembered and answered as soon as verification passes. Thresholds are set in configuration.

1. The bot greets the caller and asks for their registered mobile number (or the customer is already known).
2. The caller speaks. Speech-to-text writes down what they said; the same recording is the voice sample.
3. Voice ID compares the voice with the enrolled voiceprint. The score is above the accept threshold (0.55), so the caller is verified.
4. The bot detects the intent: balance, due date, or both.
5. Code reads the latest statement for the card: outstanding balance, minimum due, due date, overdue flag.
6. Code writes a correct draft answer. The LLM rewords it to sound natural, and a checker confirms that every amount and date matches the database.

7 The bot speaks the answer, reading amounts naturally ("four thousand two hundred and fifty dirhams and seventy-five fils"), and asks "Is there anything else I can help with?"

8 The caller says goodbye. The call ends with outcome **served**.

5. Alternative flows

Situation	What the bot does	Outcome
Voice score between 0.40 and 0.55	Step-up: asks for the card's last 4 digits and date of birth. Correct answers verify the caller.	Served
Voice score below 0.40	Says nothing about the account. Offers a retry or an agent.	Retry
3 failed verifications	Locks the customer for a set time and hands over. New calls for a locked customer go straight to an agent.	Failed verification
Typed input or no voiceprint	No voice to check, so goes straight to the step-up check.	Served
Mobile number not found 3 times	Hands over without identifying anyone.	Handed over
Customer has two cards	Asks which card (by last 4 digits) and answers for that card.	Served
Payment overdue / card blocked	Answer must say "overdue" or "blocked"; the checker enforces it.	Served
Zero balance	Says there is no outstanding balance and no payment due.	Served
Out-of-scope request	Explains what it can do and offers an agent; "yes" transfers.	Handed over
Caller asks for an agent	Transfers at any point, with reason code AGENT_REQUESTED.	Handed over
LLM down or wrong	Uses keyword rules for intent and speaks the code's draft answer.	Served

6. Design principle: code for facts, LLM for language

The LLM never invents, computes or chooses a number. It only understands the request and makes the reply sound natural. Everything that must be correct is done by code.

Task	Done by
Read balance, minimum due, due date, overdue flag	Code (SQL); money stored as whole fils, no floats
Format amounts (AED 4,250.75) and dates (25 November)	Code
Write a correct draft answer	Code
Classify intent (CARD_BALANCE, DUE_DATE, BOTH, OTHER, AGENT, REPEAT, GOODBYE)	LLM with JSON-schema output, validated by Pydantic
Rephrase the draft as a short, polite spoken sentence	LLM
Check every amount and date before speaking	Code: wrong, missing or extra numbers mean the draft is used
Decide whether the caller is verified	Code only; the LLM never sees verification answers

7. Security and privacy guardrails

- **No data before verification.** The code that reads account data refuses to run unless the call is verified, and an automated test walks every pre-verification step to prove nothing leaks.
- **Voice ID is not enough on its own.** Recordings can be replayed and voices can be cloned. The step-up check stays in place; a real deployment would add anti-spoofing (liveness) detection, caller-line and device signals, and a one-time password for anything sensitive.
- **Never spoken:** the full card number, CVV or full date of birth. Only the last 4 card digits.
- **Rate limiting:** 3 failed verifications lock the customer and hand over the call.
- **PII in logs:** mobile and card numbers masked, dates with a year replaced, identification and step-up answers never stored. Mobile numbers are kept only as a salted hash plus a masked copy.
- **The LLM is not a security boundary:** a prompt-injection phrase changed one intent label in testing, but it cannot skip verification or change the numbers spoken.
- **Audit trail:** every turn logs the masked transcript, intent, voice score, model, prompt version and latency per stage.

8. Solution and technology

Component	Technology (all local, no paid APIs)
Speech to text	faster-whisper (model size configurable, default "small", CPU int8)
Text to speech	Piper TTS, English voice en_US-lessac-medium
Voice ID	SpeechBrain ECAPA-TDNN (spkrec-ecapa-voxceleb), cosine similarity, two thresholds
Intent and phrasing	Ollama running llama3.1:8b, JSON output validated by Pydantic
Application	Python, FastAPI, call state machine; web pages with push-to-talk
Data	PostgreSQL 16 in Docker; 20 synthetic customers incl. edge cases
Testing	pytest with fake speech and LLM providers (no models or microphone needed)

9. Demo results (measured on a laptop, CPU only)

Check	Result
Genuine caller (enrolled voice)	Voice score 0.78 - 0.88: verified, correct answer spoken
Impostor (pitch-shifted voice)	Voice score 0.18 - 0.21: rejected, no account data spoken
Caller says their mobile number	Same utterance identified and verified the customer (0.83 - 0.85)
Intent detection on sample phrases	8 of 9 correct (the miss was a prompt-injection probe)
LLM answers passing the checker	12 of 12 (standard, overdue, blocked, zero-balance cards)
Time per voice turn	About 1.5 - 3 seconds end to end
Automated tests	60 passing

These are smoke tests using a synthetic Piper voice as the caller, not a statistically valid evaluation. Thresholds must be tuned with real enrolled speakers and microphones.

10. Proposed KPIs for a pilot

KPI	What it shows
Containment rate	Share of balance calls fully served without an agent
Voice ID pass rate / step-up rate	How often genuine callers get through on voice alone
False accept rate	Impostors accepted by Voice ID (measured in controlled tests)
Handover reasons	Which reason codes drive agent load (out of scope, locked, not identified)
Checker rejection rate	How often the LLM's wording was replaced by the draft
Average call time and latency	Customer effort and system speed per turn

11. Next steps

- Add Arabic: bot phrases, prompts, an Arabic Piper voice and Arabic number reading.
- Add anti-spoofing / liveness detection and a one-time-password step-up before any real use.
- Tune thresholds with real enrolled voices; measure false accept and false reject rates.
- Connect to a telephony channel (SIP / contact-centre platform) instead of the web page.
- Replace the in-memory call state with a shared store so the service can scale out.